

# PolyForge: Joint Cross-Layer Control of Replicas, Semantic Cache, and Model Tier for Multi-Tenant Hybrid Serving

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# The problem

- Multi-tenant platforms serve **hybrid** workloads: CRUD + AI inference, same tenants, same cluster.
- Three control surfaces set cost and quality: **replica autoscaling**, **semantic caching**, **model-tier routing**.
- Today: three independent controllers, each locally optimal, none aware of the others — yet they share one budget, one SLO, one tenant set.

**Question: do per-layer local optima compose?**

# Thesis statement

*Per-layer local optima do **not** compose. A controller that plans replicas, cache budget, and model tier **together** — one receding-horizon optimization against  $J = \text{cost} + 2 \cdot \text{violation} + 0.5(1 - \text{Jain})$  — dominates tuned per-layer baselines.*

Tested under **13 pre-registered protocols** (pushed before runs), **tuned** baselines, paired seeds, effect sizes with bootstrap CIs — and **published nulls**.

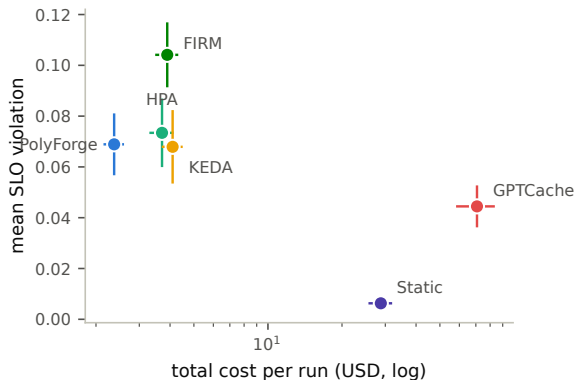
## PolyForge in one loop

- **Control plane** (Go): tenants, API keys, RLS-proven isolation, telemetry.
- **AI gateway** (Go): routing, semantic cache with cost-aware eviction.
- **Classifier**: 12-feature windows  $\rightarrow$  5-class taxonomy, drift-checked, every 10 s.
- **JCAC planner**: MPC over a move-blocked lattice ( $\leq 60$  candidates/tenant); the deployed service **imports the simulator's controller** — same code offline and online.
- **Operator**: 4 CRDs; guardrails — replica bounds, budget caps, last-good-plan fallback; shared-target plans are summed, not overwritten.

## Methodology: the part built to be attacked

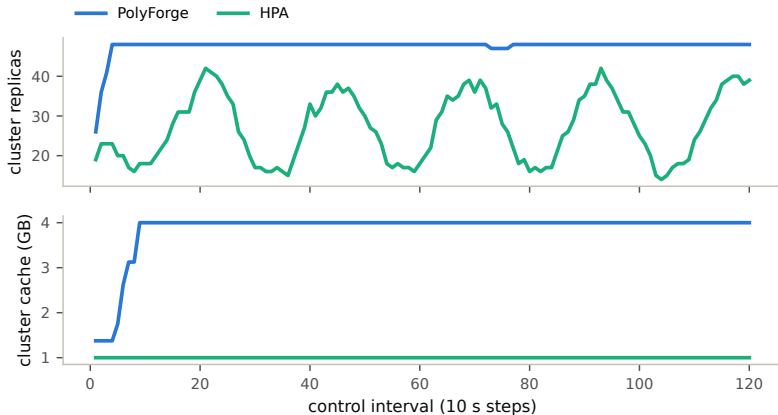
- 13 pre-registrations committed **and pushed before any run**; closed campaigns immutable; stopping rules enforced.
- Baselines (HPA, KEDA, FIRM, GPTCache-LRU, static, VTC-replica, iso-cost) **grid-tuned on our own objective**.
- Paired seeds; citable unit =  $d_z$  with 95% bootstrap CI; p-values appear once, as preregates.
- Substrates never mixed; samples never pooled; p95 not p99 (documented).
- **Honest-nulls ledger**: 8 published negatives.

## Headline matrix: 1,800 runs, 500 ablations



Wins composite  $J$  vs. every tuned baseline ( $|d_z|$  0.66–1.44); only Pareto-undominated system.  
Ablations: –jointness = +2884% cost.

## Joint control, visible in one trace



First minute of an agentic burst: replicas *and* cache move together; HPA has one knob.

## Real demand: BurstGPT replays

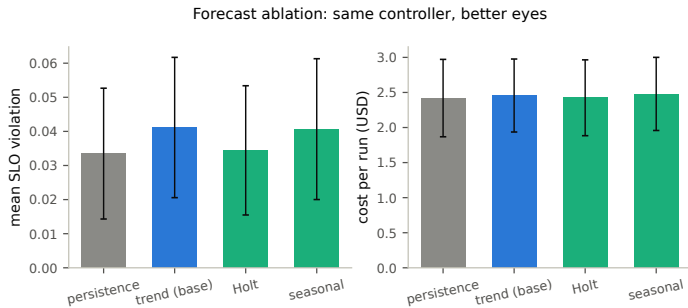
- Real trace: 10.63M requests / 335 days, deterministic ETL.
- First sample  $n=16$ : **HT FAIL** (underpowered) — point estimates reproduce the v1 ranking. Published as measured.
- Powered replay  $n=96$  (new prereg): **HT2 PASS** —  $J$  beats HPA/KEDA/FIRM ( $p \leq 5.5 \times 10^{-5}$ ,  $d_z -0.43 \dots -0.47$ ), **cost -70%** ( $p \leq 4.3 \times 10^{-6}$ ), violation trade +0.07 **disclosed as pre-registered**.

Cite -70% ( $n=96$ ), never the underpowered sample's -76%.

# The SLO story, told honestly

- v2 H1: beat tuned HPA/KEDA on violation? **FAIL** — parity, at  $-43 \dots -48\%$  cost.
- v3 H1' (overload cells): **FAIL** — reactive scalers buy attainment at **3.2–3.4 $\times$  spend**; only PolyForge honors the capacity cap.
- v3 H2': seasonal  $<$  trend on violation where periodicity exists: **PASS** ( $p = 6 \times 10^{-4}$ , no cost premium).
- Spike class (declared exploratory): beats HPA on violation *and* cost ( $p = 5 \times 10^{-5}$ ,  $-46\%$ ).
- Stopping rule: no third confirmatory attempt. The goalposts are in the git history.

## Forecasting: the data corrected us



Synthetic stand-in predicted seasonal —44.8% RMSE — **did not transfer** to real BurstGPT (Holt —26.9% wins). Azure LLM 2024 reproduces the boundary: the winner tracks measured periodicity  $\Rightarrow$  per-tenant, pluggable forecasting.

## Semantic cache on real LMSYS-Chat-1M

- Adaptive hit rate **29.6% @ cosine 0.85** (48.8% @ 0.70); +93% vs. a 10%-fixed cache; \$0.296/1k queries.
- Empirical  $h(K)$ :  $h_{\max}=0.285$ ,  $K_{1/2}\approx 6.6\text{k}$  entries — sim's curve was optimistic; documented, constants untouched.
- **Hit precision measured** (frozen protocol): 0.313 overall @  $\tau=0.85$ ; same-model  $\approx 2\times$  cross-model; near-duplicate ceiling 0.338  $\Rightarrow$  cite the shape, not the absolute.

# Fairness and security

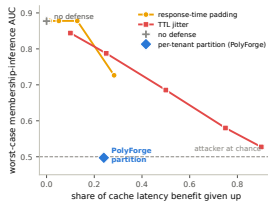
## Fairness (control-plane altitude)

- vs. FIRM: Jain 0.9689 / 0.9294 ( $d_z=1.0$ ).
- vs. tuned VTC-replica: wins  $J$  ( $d_z=-1.12$ ) and *Jain itself* (0.9705 / 0.9599) at  $-35\%$  cost.
- Own  $\gamma$ -term: **published null**.

## Security

- Shared cache leaks membership: AUC **0.88** from timing alone.
- Per-tenant partitioning  $\rightarrow$  chance (0.50) at 24% latency cost, 76% hits kept — dominates padding & TTL jitter.  
**Confirmed over the wire** (real gateway, AUC 0.502).

Defense frontier: partitioning dominates the known mitigations



## Theory, at the code's altitude

- **Prop. 1:** the two-sweep solver returns a block-coordinate minimum of  $J$  — no tenant improves unilaterally. (Not claimed: global optimality; Jain coupling is non-separable.)
- **Prop. 2:** self-calibration scale is confined to  $[0.5, 1]$ , only ever conservative, and returns to full trust in  $\leq 25$  agreeing steps — safety independent of tuning.
- Measured scale: planner growth exponent **1.45**; 3 s timeout crossed at 128 tenants; past it, designed fallback.

## Validity: what bounds the simulator gap

- **Calibration errs against us:** congestion fit  $a=0.86$  vs. assumed 1.0; GPU tier bench confirms ordering (1:1.5:16.6).
- **Real data carries the claims:** BurstGPT + Azure demand, LMSYS prompts.
- **Live check executed, ordinal-only:** both arms ran on a real kind cluster (12/12 valid); the pre-frozen protocol recorded a **DISAGREE** — live *parity* where the sim separates the arms (cache/tier knobs inert live by construction). Published as measured; no claim rested on live-ranking transfer.

Not claimed: absolute dollars/latency on production fleets; token-level fairness; p99 on sim.

# Contributions

- 1 First joint replicas+cache+tier controller with per-tenant dollar accounting.
- 2 The composition gap, measured: +2884% cost without jointness; -70% cost on real demand vs. tuned scalers.
- 3 Honest SLO account: parity at -43...-48% cost; mechanism confirmed; two failures published.
- 4 Real-data cache results *with a quality denominator*.
- 5 Fairness wins at stated altitude; own term nulled in public.
- 6 Cache side-channel attack + dominant defense frontier, confirmed over the wire.
- 7 Two construction-level solver guarantees.
- 8 A pre-registered, null-publishing systems methodology — in git, diffable.

## What remains, and closing

**Done since the proposal:** the live JCAC ordinal slice ran (12/12, DISAGREE published); tier-price and empirical-cache-curve reruns (both PASS 5/5); sim chaos (a dead planner still beats a healthy HPA); planner cells to **1024 tenants**; the isolation attack executed over the wire.

**Remaining (user-gated, GPU host):** the **three-knob live plane** — the joint controller with cache *and* tier as live levers, not just replicas (pre-registered as the Wave 4 live plane); plus live chaos and a live p99 number.

*The strongest sentence in this thesis is not a percentage — it is that every percentage traces to a protocol committed before the run.*